

JavaScript Assignment

Task 1: Array Operations

Description

Create a JavaScript program that performs multiple operations on an array of numbers entered by the user.

Requirements

1. Ask the user to enter the **size of the array**.
2. Receive array values from the user.
3. Validate that **all entered values are numbers**.
4. Ask the user to select one of the following options:

Letter	Operation
a	Display the array in the same order it was entered
b	Display the array in ascending order
c	Display the array in descending order
d	Display the reversed version of the original array
e	Display only even numbers from the array. If none exist, show an alert
f	Ask the user for a number, then display all array elements divisible by it. If none exist, show an alert
g	Display a new array where all numbers have a 30% discount applied
h	Display a string of all numbers concatenated using ***
5.	After completing any selected operation, ask the user if they want to repeat the program .

Task 2: Count Vowels

Description

Write a function that:

- Takes a string as input
 - Returns the number of vowels (a, e, i, o, u) in the string
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Task 3: Capitalize Words

Description

Write a function that:

- Takes a string as input
 - Returns a new string where the **first letter of each word is capitalized**
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Task 4: Count Character Occurrences

Description

Write a function that:

- Takes a string and a character as input
 - Returns the number of times the character appears in the string
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Task 5: Count Words

Description

Write a function that:

- Takes a string as input

- Returns the number of words in the string
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Task 6: User Name and Mobile Validation

Description

1. Ask the user to enter their **name**
 - Ensure the input is a **string**
 - Re-prompt until a valid string is entered
 2. Ask the user to enter their **mobile number**
 - Must follow this format: **xxx-xxxxxxx**
 - Must be **11 characters long**
 - Must start with one of the following:
 - **010**
 - **011**
 - **012**
 - **015**
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Task 7: Date Object & Validation

Part 1: Research

- Search and study the **JavaScript Date object**
 - Understand its constructors and common methods
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Part 2: Birth Date Validation

1. Prompt the user to enter their birth date in the following format:

DD-MM-YYYY

Example: 22-01-1999

2. Create a function that:

- Accepts the date string as a parameter
- Ensures:
 - The string length is exactly **10 characters**
 - Contains - at index **2** and **5**

a. Valid Format

- Create a `Date` object using:
`Date(year, month, day)`
- Display the date using `toString()` in an alert

b. Invalid Format

- Display an alert:
"Wrong Date Format"

Additional Requirements

1. Create a function that:
 - Accepts a date string
 - Returns the **day name** (Saturday, Sunday, ...)
2. Modify **Task 6**:
 - Ask the user to enter their mobile number in **international format**
 - Format: 00201xxxxxxxxx
 - Validation rules:
 - Exactly **14 digits**

- Numbers only
- Starts with 002
- Followed by 010, 011, or 012
- Remaining 8 digits

3. Modify **Task 2**:

- Calculate the user's **age in years, months, and days**
- Handle edge cases correctly
Example:
 - Birth date: 30-12-2000
 - Current date: 01-11-2018
 - Result must **not contain negative values**